# Chapter 3. Connectedness and Compactness

Connected and path-connected spaces, components, compact and limit point compact spaces, local compactness

## Contents

## 3.1 Connected spaces

**Motivation.** The intermediate value theorem is really a statement about the shape of $[a, b]$: it cannot be split into two disjoint open pieces. Connectedness abstracts this, and continuous images of connected spaces are connected, which gives the intermediate value theorem in any setting.

A **separation** of $X$ is a pair $U, V$ of disjoint nonempty open sets with $X = U \cup V$. $X$ is **connected** if no separation exists; equivalently, the only sets both open and closed are $\emptyset$ and $X$.

**Lemma 3.1 (23.1)**

If $Y \subset X$, a separation of $Y$ is a pair of disjoint nonempty sets $A, B$ with $A \cup B = Y$, neither containing a limit point of the other.

**Proof.** If $A, B$ separate $Y$, $A$ is closed in $Y$, so $A = \overline{A} \cap Y$ (Theorem 2.14) and $\overline{A} \cap B = \emptyset$: $B$ contains no limit point of $A$; symmetrically. Conversely, if $\overline{A} \cap B = \emptyset = A \cap \overline{B}$, then $\overline{A} \cap Y = A$ and $\overline{B} \cap Y = B$ are closed in $Y$, so their complements $B$, $A$ are open in $Y$. ■

Examples: the indiscrete topology is connected; $\mathbb{Q}$ is totally disconnected (its only connected subspaces are points); $Y = [-1, 0) \cup (0, 1]$ is disconnected.

**Lemma 3.2 (23.2)**

If $C, D$ separate $X$ and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

**Proof.** $C \cap Y$ and $D \cap Y$ are open in $Y$ and disjoint with union $Y$; if both were nonempty they would separate $Y$. ■

**Theorem 3.3 (23.3)**

The union of a collection of connected subspaces of $X$ with a point in common is connected.

**Proof.** Let $p$ be the common point and suppose $C, D$ separate $Y = \bigcup A_\alpha$, with $p \in C$. Each $A_\alpha$ lies in $C$ or $D$ (Lemma **??**), hence in $C$ as $p \in A_\alpha$. So $D = \emptyset$, a contradiction. ■

**Theorem 3.4 (23.4)**

If $A$ is connected and $A \subset B \subset \overline{A}$, then $B$ is connected.

**Proof.** If $C, D$ separate $B$, then $A \subset C$, say (Lemma **??**); then $\overline{A} \subset \overline{C}$, and $\overline{C} \cap D = \emptyset$ (Lemma **??**), so $B \cap D \subset \overline{A} \cap D = \emptyset$. ■

**Theorem 3.5 (23.5)**

The image of a connected space under a continuous map is connected.

**Proof.** If $A, B$ separate $f(X)$, then $f^{-1}(A), f^{-1}(B)$ separate $X$. ■

**Theorem 3.6 (23.6)**

A finite cartesian product of connected spaces is connected.

**Proof.** For $X \times Y$ fix $a \times b$. For each $x$, $T_x = (X \times b) \cup (x \times Y)$ is connected (Theorem **??**, with common point $x \times b$). All $T_x$ contain $a \times b$, and their union is $X \times Y$. Induct for more factors. ■

Arbitrary products are connected in the product topology (the set of points differing from a fixed point in finitely many coordinates is a connected dense subset), but $\mathbb{R}^\omega$ in the box topology is not: bounded and unbounded sequences form a separation.

## 3.2 Connected subspaces of the real line

**Motivation.** The real line is connected because of the least upper bound property; from this we get that intervals are connected and, through path connectedness, that $\mathbb{R}^n$ and spheres are connected.

A simply ordered set $L$ with more than one element is a **linear continuum** if it has the least upper bound property and for $x < y$ there is $z$ with $x < z < y$.

**Theorem 3.7 (24.1)**

A linear continuum $L$ in the order topology is connected, and so are its intervals and rays.

**Proof.** It suffices to show convex subsets $Y$ are connected. Suppose $A, B$ separate $Y$, $a \in A$, $b \in B$, $a < b$. $[a, b] \subset Y$, so $A_0 = A \cap [a, b]$, $B_0 = B \cap [a, b]$ separate $[a, b]$. Let $c = \sup A_0$. If $c \in B_0$, then $c \neq a$, so $c > a$; $B_0$ is open in $[a, b]$, so some $(d, c] \subset B_0$, and then $d$ is a smaller upper bound for $A_0$ (since $(d, c]$ misses $A_0$ and $c$ is the sup): contradiction. If $c \in A_0$, then $c < b$ and some $[c, e) \subset A_0$; choosing $z$ with $c < z < e$ gives $z \in A_0$, contradicting $c = \sup A_0$. ■

**Corollary 3.8 (24.2)**

$\mathbb{R}$ is connected, and so are its intervals and rays.

**Theorem 3.9 (24.3, intermediate value theorem)**

Let $f : X \to Y$ be continuous, $X$ connected, $Y$ ordered in the order topology. If $a, b \in X$ and $r$ lies between $f(a)$ and $f(b)$, then $f(c) = r$ for some $c \in X$.

**Proof.** Otherwise $f(X) \cap (-\infty, r)$ and $f(X) \cap (r, +\infty)$ separate the connected set $f(X)$. ■

Other linear continua: $I \times I$ in the dictionary order (the ordered square), and the **long line** $L = S_\Omega \times [0, 1)$. A **path** from $x$ to $y$ is a continuous $f : [a, b] \to X$ with $f(a) = x$, $f(b) = y$; $X$ is **path connected** if every two points are joined by a path. Path connected spaces are connected (Theorems **??**, **??**, **??**).

**Example 3.1**

(a) The unit ball $B^n$ and punctured space $\mathbb{R}^n - \{0\}$ $(n > 1)$ are path connected, hence so is $S^{n-1}$ (its image under $x \mapsto x/\|x\|$).
(b) The ordered square is connected but not path connected: a path from the least to the greatest element would be surjective, and the uncountably many disjoint open intervals $U_x = (x \times 0, x \times 1)$ would have disjoint open preimages each containing a rational.
(c) The **topologist's sine curve** $\overline{S}$, the closure in $\mathbb{R}^2$ of $S = \{(x, \sin(1/x)) : 0 < x \leq 1\}$, is connected (Theorem **??**) but not path connected: if $f : [a, c] \to \overline{S}$ were a path from $0 \times 0$ into $S$, let $b = \sup\{t : f(t) \in 0 \times [-1, 1]\}$; on $(b, c]$, $f(t) \in S$, and choosing $t_n \downarrow b$ with $\sin(1/x(t_n)) = (-1)^n$ contradicts the continuity of the second coordinate at $b$.

## 3.3 Components and local connectedness

**Motivation.** A disconnected space breaks into maximal connected pieces, its components. Counting components is the simplest topological invariant, used for instance to show that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

Define $x \sim y$ if some connected subspace contains both; the classes are the **components** of $X$.

**Theorem 3.10 (25.1)**

The components of $X$ are connected disjoint subspaces whose union is $X$, and each nonempty connected subspace meets only one of them.

**Proof.** $\sim$ is an equivalence relation (transitivity by Theorem **??**). A connected $A$ meeting components $C_1, C_2$ in $x_1, x_2$ gives $x_1 \sim x_2$. For connectedness of a component $C$, fix $x_0 \in C$; each $x \in C$ lies in a connected $A_x \ni x_0$, contained in $C$ by the previous sentence, and $C = \bigcup A_x$ is connected by Theorem **??**. ■

Similarly, define $x \sim y$ if a path joins them; the classes are the **path components**.

**Theorem 3.11 (25.2)**

The path components of $X$ are path-connected disjoint subspaces with union $X$, and each nonempty path-connected subspace meets only one of them.

**Proof.** Transitivity: concatenate paths $f : [a, b] \to X$ and $g : [b, c] \to X$ via the pasting lemma. ■

Components are closed (the closure of a connected set is connected); path components need not be (the topologist's sine curve has two path components, $S$ and $0 \times [-1, 1]$, the first not closed). $X$ is **locally connected at** $x$ if every neighbourhood of $x$ contains a connected neighbourhood of $x$; similarly **locally path connected**. Intervals in $\mathbb{R}$ are both; $\mathbb{Q}$ is neither; the topologist's sine curve is connected but not locally connected.

**Theorem 3.12 (25.3)**

$X$ is locally connected iff for every open $U$ each component of $U$ is open in $X$.

**Proof.** ($\Rightarrow$) If $C$ is a component of $U$ and $x \in C$, a connected neighbourhood $V \subset U$ of $x$ lies in $C$.

($\Leftarrow$) For a neighbourhood $U$ of $x$, the component of $U$ containing $x$ is a connected neighbourhood inside $U$. ■

**Theorem 3.13 (25.4)**

$X$ is locally path connected iff for every open $U$ each path component of $U$ is open in $X$.

**Theorem 3.14 (25.5)**

Each path component of $X$ lies in a component. If $X$ is locally path connected, components and path components coincide.

**Proof.** The first claim since path connected sets are connected. Let $C$ be a component and $P \subset C$ a path component. The path components of $X$ are open (Theorem **??** with $U = X$); $Q$, the union of the other path components meeting $C$, is open, so $P, Q$ would separate $C$ unless $Q = \emptyset$. Hence $P = C$. ■

## 3.4 Compact spaces

**Motivation.** On $[a, b]$ continuous functions are bounded and attain their extrema, and they are uniformly continuous. The property responsible is compactness: every open cover has a finite subcover. It lets us pass from local information to global information, and it is preserved by continuous maps and products.

A collection of subsets **covers** $X$ if its union is $X$; it is an **open covering** if its members are open. $X$ is **compact** if every open covering contains a finite subcollection covering $X$. Examples: $\mathbb{R}$ is not compact; $\{0\} \cup \{1/n\}$ is compact; finite spaces and spaces with the finite complement topology are compact.

**Lemma 3.15 (26.1)**

A subspace $Y \subset X$ is compact iff every covering of $Y$ by sets open in $X$ contains a finite subcollection covering $Y$.

**Proof.** Open sets of $Y$ are intersections $Y \cap A$ with $A$ open in $X$; translate coverings in both directions. ■

**Theorem 3.16 (26.2)**

Every closed subspace of a compact space is compact.

**Proof.** Given a covering $\mathcal{A}$ of a closed $Y$ by sets open in $X$, $\mathcal{A} \cup \{X - Y\}$ covers $X$; a finite subcover, minus $X - Y$, covers $Y$. ■

**Theorem 3.17 (26.3)**

Every compact subspace of a Hausdorff space is closed.

**Lemma 3.18 (26.4)**

If $Y$ is a compact subspace of a Hausdorff space $X$ and $x_0 \notin Y$, there are disjoint open $U \ni x_0$ and $V \supset Y$.

**Proof.** For each $y \in Y$ choose disjoint neighbourhoods $U_y \ni x_0$, $V_y \ni y$. Finitely many $V_{y_1}, \ldots, V_{y_n}$ cover $Y$; take $V = \bigcup V_{y_i}$ and $U = \bigcap U_{y_i}$. This proves Theorem **??** too: $X - Y$ is open. ■

**Theorem 3.19 (26.5)**

The image of a compact space under a continuous map is compact.

**Proof.** If $\mathcal{A}$ covers $f(X)$ by open sets of $Y$, $\{f^{-1}(A)\}$ covers $X$; finitely many suffice. ■

**Theorem 3.20 (26.6)**

A continuous bijection $f : X \to Y$ from a compact space to a Hausdorff space is a homeomorphism.

**Proof.** For closed $A \subset X$: $A$ is compact (Theorem **??**), $f(A)$ is compact (**??**), hence closed (**??**). So $f^{-1}$ is continuous. ■

**Lemma 3.21 (26.8, the tube lemma)**

If $Y$ is compact and $N$ is an open set of $X \times Y$ containing the slice $x_0 \times Y$, then $N$ contains a tube $W \times Y$ with $W$ a neighbourhood of $x_0$.

**Proof.** Cover $x_0 \times Y$ by basis elements $U \times V \subset N$ meeting it; it is compact (homeomorphic to $Y$), so finitely many $U_i \times V_i$ cover it. Let $W = \bigcap U_i$. Every $x \times y$ with $x \in W$ lies in some $U_i \times V_i$ (choose $i$ with $y \in V_i$), so $W \times Y \subset N$. ■

**Theorem 3.22 (26.7)**

A product of finitely many compact spaces is compact.

**Proof.** Let $\mathcal{A}$ be an open cover of $X \times Y$. For each $x_0$, finitely many members cover $x_0 \times Y$; their union $N$ contains a tube $W_{x_0} \times Y$. Finitely many $W_{x_i}$ cover $X$, and the corresponding finitely many members of $\mathcal{A}$ cover $X \times Y$. ■

A collection $\mathcal{C}$ has the **finite intersection property** if every finite subcollection has nonempty intersection.

**Theorem 3.23 (26.9)**

$X$ is compact iff every collection $\mathcal{C}$ of closed sets with the finite intersection property has $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.

**Proof.** Pass to complements: $\{X - C\}$ is an open cover iff $\bigcap C = \emptyset$, and a finite subcollection covers iff the corresponding finite intersection is empty. ■

In particular nested sequences $C_1 \supset C_2 \supset \cdots$ of nonempty closed sets in a compact space have nonempty intersection.

## 3.5 Compact subspaces of the real line

**Motivation.** We now prove that $[a, b]$ really is compact, and hence that closed bounded sets in $\mathbb{R}^n$ are compact. The extreme value theorem, the Lebesgue number lemma and uniform continuity all follow.

**Theorem 3.24 (27.1)**

If $X$ is simply ordered with the least upper bound property, every closed interval $[a, b]$ in $X$ is compact (order topology).

**Proof.** Let $\mathcal{A}$ be a covering of $[a, b]$ by sets open in $[a, b]$.

**Step 1.** for $x \in [a, b)$ there is $y > x$ such that $[x, y]$ is covered by at most two members of $\mathcal{A}$ (if $x$ has an immediate successor $y$, use the members containing $x, y$; otherwise take $A \ni x$ containing $[x, c)$ and any $y \in (x, c)$).

**Step 2.** let $C$ be the set of $y > a$ in $[a, b]$ such that $[a, y]$ is finitely covered; $C \neq \emptyset$ by Step 1. Let $c = \sup C$.

**Step 3.** $c \in C$: choose $A \ni c$ containing some $(d, c]$; if $c \notin C$, some $z \in C \cap (d, c)$, and $[a, z]$ plus $A$ covers $[a, c]$.

**Step 4.** $c = b$: otherwise Step 1 extends past $c$. ■

**Corollary 3.25 (27.2)**

Every closed interval in $\mathbb{R}$ is compact.

**Theorem 3.26 (27.3)**

$A \subset \mathbb{R}^n$ is compact iff it is closed and bounded (in the euclidean or square metric).

**Proof.** If $A$ is compact, it is closed (Theorem **??**) and covered by finitely many open balls $B_\rho(0, m)$, hence bounded. Conversely, a closed bounded $A$ lies in a cube $[-P, P]^n$, compact by Theorems **??** and **??**; apply Theorem **??**. ■

**Theorem 3.27 (27.4, extreme value theorem)**

If $f : X \to Y$ is continuous, $X$ compact and $Y$ ordered, there are $c, d$ with $f(c) \leq f(x) \leq f(d)$ for all $x$.

**Proof.** $A = f(X)$ is compact. If $A$ had no largest element, the rays $(-\infty, a)$, $a \in A$, would cover $A$; a finite subcover $(-\infty, a_i)$ misses the largest $a_i$. So $A$ has a largest element $f(d)$; similarly a smallest. ■

For $A \subset X$ in a metric space, $d(x, A) = \inf_{a \in A} d(x, a)$ is continuous in $x$ (indeed $|d(x, A) - d(y, A)| \leq d(x, y)$).

**Lemma 3.28 (27.5, Lebesgue number lemma)**

Let $\mathcal{A}$ be an open covering of a compact metric space $X$. There is $\delta > 0$ such that every subset of diameter less than $\delta$ lies in some member of $\mathcal{A}$.

**Proof.** If $X \in \mathcal{A}$ take any $\delta$. Otherwise choose a finite subcover $A_1, \ldots, A_n$, let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$. $f > 0$ (each $x$ lies in some $A_i$, so $d(x, C_i) > 0$) and continuous, so $f \geq \delta > 0$ (Theorem **??**). If $\operatorname{diam} B < \delta$ and $x_0 \in B$, then $B \subset B(x_0, \delta)$; choosing $m$ with $d(x_0, C_m)$ maximal gives $d(x_0, C_m) \geq f(x_0) \geq \delta$, so $B(x_0, \delta) \subset A_m$. ■

**Theorem 3.29 (27.6, uniform continuity theorem)**

A continuous map from a compact metric space to a metric space is uniformly continuous.

**Proof.** Given $\epsilon$, cover $X$ by the open sets $f^{-1}(B(y, \epsilon/2))$ and let $\delta$ be a Lebesgue number. If $d(x_1, x_2) < \delta$, $\{x_1, x_2\}$ lies in one of these sets, so $d(f(x_1), f(x_2)) < \epsilon$. ■

A point $x$ is **isolated** if $\{x\}$ is open.

**Theorem 3.30 (27.7)**

A nonempty compact Hausdorff space without isolated points is uncountable.

**Proof.**

**Step 1.** for any nonempty open $U$ and point $x$, there is a nonempty open $V \subset U$ with $x \notin \overline{V}$ (choose $y \in U$, $y \neq x$, which exists since $U \neq \{x\}$; separate $x$ and $y$ by disjoint $W_1 \ni x$, $W_2 \ni y$; $V = W_2 \cap U$).

**Step 2.** given $f : \mathbb{Z}_+ \to X$, apply Step 1 recursively to get nonempty open $V_1 \supset V_2 \supset \cdots$ with $f(n) \notin \overline{V_n}$ and $\overline{V_{n+1}} \subset V_n$. The nested closed sets $\overline{V_n}$ have a common point $x$ (Theorem **??**), and $x \neq f(n)$ for all $n$. ■

**Corollary 3.31 (27.8)**

Every closed interval in $\mathbb{R}$ is uncountable.

## 3.6 Limit point compactness

**Motivation.** In analysis compactness is often defined with sequences (Bolzano–Weierstrass). We compare the cover, limit-point and sequential versions and show that they agree for metric spaces.

$X$ is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

**Theorem 3.32 (28.1)**

Compactness implies limit point compactness, but not conversely.

**Proof.** If $A$ has no limit point, $A$ is closed and hence compact, and each $a \in A$ has a neighbourhood $U_a$ with $U_a \cap A = \{a\}$. The $U_a$ cover the compact $A$, finitely many suffice, and each contains one point of $A$: $A$ is finite. Counterexample to the converse: $X = \mathbb{Z}_+ \times Y$ with $Y$ a two-point indiscrete space (every point has a limit point), or $S_\Omega$ below. ■

**Theorem 3.33 (28.2)**

For a metrizable space $X$ the following are equivalent:
(1) compact;
(2) limit point compact;
(3) sequentially compact.

**Proof.** (1)⇒(2) is Theorem **??**. (2)⇒(3): if the range of $(x_n)$ is finite, a constant subsequence converges; otherwise it has a limit point $x$, and choosing $x_{n_k} \in B(x, 1/k)$ with $n_k$ increasing (possible since each ball contains infinitely many points of the range, Theorem 2.19) gives a convergent subsequence. (3)⇒(1): first, the Lebesgue number lemma holds: if not, there are $C_n$ of diameter $< 1/n$ in no member of $\mathcal{A}$; pick $x_n \in C_n$ and a convergent subsequence $x_{n_i} \to a$; $a \in A \in \mathcal{A}$ with $B(a, \epsilon) \subset A$, and for large $i$, $C_{n_i} \subset B(a, \epsilon)$, a contradiction. Second, $X$ is **totally bounded**: if some $\epsilon$ admits no finite cover by $\epsilon$-balls, choose $x_{n+1} \notin \bigcup_{i \le n} B(x_i, \epsilon)$; the sequence has no convergent subsequence since all distances are $\ge \epsilon$. Finally, with $\delta$ a Lebesgue number, cover $X$ by finitely many $\delta/3$-balls; each lies in a member of $\mathcal{A}$. ■

**Example 3.2**

$S_\Omega$ is limit point compact but not compact: an infinite subset contains a countably infinite subset $B$, which has an upper bound $b$ (Theorem 1.27), so $B \subset [a_0, b]$, which is compact (Theorem **??**); hence $B$ has a limit point. $S_\Omega$ is not compact since the sections $S_\alpha$ (or the open intervals $[a_0, \alpha)$) cover it without a finite subcover. It is also sequentially compact.

## 3.7 Local compactness

**Motivation.** $\mathbb{R}^n$ is not compact, but it is compact near every point, and adding one point at infinity makes it compact ($\mathbb{R}^2 \cup \{\infty\}$ is the sphere). Local compactness is exactly the condition for this one-point compactification to be Hausdorff.

$X$ is **locally compact at** $x$ if some compact $C$ contains a neighbourhood of $x$, and locally compact if this holds at every point. $\mathbb{R}$, $\mathbb{R}^n$, every simply ordered set with the least upper bound property are locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

**Theorem 3.34 (29.1)**

$X$ is locally compact Hausdorff iff there is a space $Y$ with:
(1) $X$ is a subspace of $Y$;
(2) $Y - X$ is a single point;
(3) $Y$ is compact Hausdorff. Two such $Y$ are homeomorphic by a homeomorphism fixing $X$.

**Proof.** Uniqueness: map $Y \to Y'$ by the identity on $X$ and $p \mapsto q$; open sets of $Y$ containing $p$ are complements of compact subsets of $X$ ($Y - U$ is closed in $Y$, hence compact, and lies in $X$), and the same holds in $Y'$.

Existence: let $Y = X \cup \{\infty\}$ with open sets the open sets of $X$ and the sets $Y - C$ for compact $C \subset X$. This is a topology (uses that compact sets in the Hausdorff $X$ are closed); $X$ is a subspace; $Y$ is compact (a cover contains some $Y - C$, and $C$ needs finitely many more); $Y$ is Hausdorff (separate $x$ from $\infty$ by $\operatorname{Int} C \ni x$ and $Y - C$, where $C$ is a compact set containing a neighbourhood of $x$). Conversely, a subspace of a compact Hausdorff space with one point removed is Hausdorff and locally compact (separate $x$ from the removed point by disjoint open $U \ni x$, $V \ni \infty$; $C = Y - V$ is compact and contains $U$). ■

$Y$ is the **one-point compactification**; e.g. $\mathbb{R} \cup \{\infty\} \cong S^1$ and $\mathbb{R}^n \cup \{\infty\} \cong S^n$ (stereographic projection).

**Theorem 3.35 (29.2)**

A Hausdorff space $X$ is locally compact iff for every $x$ and neighbourhood $U$ of $x$ there is a neighbourhood $V$ of $x$ with $\overline{V}$ compact and $\overline{V} \subset U$.

**Proof.** ($\Leftarrow$) Take $C = \overline{V}$.

($\Rightarrow$) In the one-point compactification $Y$, $C = Y - U$ is closed, hence compact; by Lemma **??** there are disjoint open $V \ni x$, $W \supset C$. Then $\overline{V}$ (closure in $Y$) misses $W$, so $\overline{V} \subset U$; and $\overline{V}$ is compact as a closed subset of $Y$. ■

**Corollary 3.36 (29.3)**

A closed or open subspace of a locally compact Hausdorff space is locally compact.

**Proof.** Closed $A$: if $C$ is compact containing a neighbourhood $U$ of $x$, then $C \cap A$ is compact and contains $U \cap A$. Open $A$: by Theorem **??**, $x \in V \subset \overline{V} \subset A$ with $\overline{V}$ compact. ■

**Corollary 3.37 (29.4)**

$X$ is homeomorphic to an open subspace of a compact Hausdorff space iff $X$ is locally compact Hausdorff.